Umusaruro

Evidence contract · Phase 1 · 15 October 2026

A number appears on a screen only when this contract names its survey, table, and year. A blank or a dash stays blank. It is not zero. Readers are a district agronomist and a MINAGRI planner. Cells were read on 15 October 2026 from the Season A 2026 PDF. A second person still checks every cell in Phase 2. Model assistance was used to draft this file. Repository: https://gitlab.com/mukunzijonathan/umusaruro

1. The product

Umusaruro is a district maize action sheet. It shows Season A 2026 maize yield against the PSTA 5 milestones, then irrigation or all-crop improved seed, and names inorganic fertilizer when that table lags. Competition deadline: 23:59 on 30 October 2026. The submission is a public Git repository, a deployed application, and documentation.

2. PSTA 5 maize milestones, tonnes per hectare

Baseline (2023)	2024/25	2025/26	2026/27	2027/28	2028/29
1.70	1.87	2.06	2.26	2.49	2.74

Season A 2026, all farms, is 1,994 kg/ha, shown as 1.99 t/ha. That is under 2.06 and under 2.26. The figure 2.26 is the 2026/27 column. Source: PSTA 5 abridged results framework. Display rule: divide kg/ha by 1,000 and round half up to two decimals.

3. Cells Season A 2026 may show

What	Table	Level
Maize yield, kg/ha	13	District, plus national rows
Large-farm yield, kg/ha	14	District. A dash stays blank.
Improved seed, all crops	20	District, by farm type
Source of improved seed	22	District
Inorganic fertilizer	25	District, by farm type
Irrigation, % of farmers	31	District, by farm type
Maize seed 64.3% improved	21	National. Method page only.
Maize loss 0.3% of use	16	National. No district store.

National maize: smallholders 1,897 kg/ha (1.90 t/ha), all farms 1,994 (1.99), large farms 4,337 (4.34). Irrigation: 14.4% overall, 12.6% of smallholders, 60.3% of large farms. All-crop improved seed, percentage of farmers: 37.0, 35.2, and 82.6. Table 21 maize seed, 64.3% improved, has no district split. Bush bean improved seed is 0.5%. Beans: 700 kg/ha nationally, 542 kg/ha on large farms. Table 20 is not the maize seed rate.

4. Three districts, and the irrigation candidates

District	Maize kg/ha	t/ha	Vs 2.26	SSF irrig.	LSF irrig.
Nyagatare	2,887	2.89	Above	6.4%	50.0%
Kirehe	2,515	2.52	Above	14.9%	78.6%
Kamonyi	1,863	1.86	Below	22.4%	93.3%

Only those two districts are at or above 2.26. Next are Rulindo 2,096 kg/ha, Muhanga 2,081, and Burera 2,042. Kamonyi large-farm maize is 6,470 kg/ha. Seed gaps, Table 20: Nyagatare 39.1 points, Kamonyi 54.7, Kirehe 36.9, Gakenke 29.2, Rwamagana 24.5, Ngoma 12.2. Fertilizer gaps, Table 25: Kamonyi 46.1, Nyagatare 23.8, Kirehe 16.9. In Gakenke, smallholders lead large farms, 85.1% against 66.7%.

Team rules, not NISR thresholds. Irrigation candidate: gap at least 30 points and smallholder irrigation under 15%, then a CHIRPS gate for September 2025–February 2026 below the 30-district median. Seed: Table 20 gap at least 25 points. If both fire, the larger gap wins unless seed leads by under 5 points. Fertilizer is named when Table 25 lags by at least 15 points. Rainfall totals are not locked here. The ten rows below are candidates, not the final action list.

Candidate	SSF %	LSF %	Gap
Kicukiro	9.9	80.0	70.1
Kayonza	12.8	78.7	65.9
Kirehe	14.9	78.6	63.7
Gatsibo	8.1	63.6	55.5
Ngoma	4.7	50.0	45.3
Nyagatare	6.4	50.0	43.6
Rwamagana	10.4	53.7	43.3
Karongi	10.8	50.0	39.2
Nyamasheke	8.8	46.2	37.4
Nyamagabe	8.8	42.9	34.1

Before rainfall: Nyagatare's action is irrigation, and fertilizer is named. Kamonyi's action is all-crop seed, and fertilizer is named; irrigation does not fire because smallholders are already at 22.4%. Kirehe's action is irrigation if rainfall passes, and fertilizer is named. Gakenke is seed only. Ngoma is irrigation, with fertilizer named; its seed gap of 12.2 does not fire. Boundaries: Kirehe 14.9 against 15, Muhanga 15.2, Gakenke's irrigation gap 28.7 against 30, Rwamagana's seed gap 24.5 against 25, Kirehe's fertilizer lag 16.9 against 15.

5. Refusals, and the Season B check

Refused: a district store; a maize variety; a satisfaction score; a learned seed-soil-fertilizer score; copying a national rate onto a district; letting Season B or Season C replace Season A; re-estimating the NST2 hectare total; letting rainfall replace a NISR cell. The 2025 plot questionnaire has a production-status reason. It is not a published district table. The Rwanda Soil Information System is a fertilizer recommendation from soil. Maize was still in testing at the November 2024 launch.

Season B, released 22 September 2026, from 1,200 segments and 505 large-scale farmers, collected 19 April to 28 June 2026. The publication page: improved seed 20.9% (smallholders 19.4%, large farms 65%); inorganic fertilizer 57.5% (56.7% and 80.2%); organic fertilizer 81.3%; irrigation 13.1%. Workbook maize yield is 1.29 t/ha, not 1.99. Season B irrigation is Table 29, and that sheet's footer still says SAS 2025. No Season B irrigation cell is shown until that footer is resolved.

6. Links

Hackathon: https://statistics.gov.rw/about/hackathon/2026-hackathon-competition
SAS Season A 2026: https://statistics.gov.rw/sites/default/files/documents/2026-04/SAS%202026%20Season%20A%20_Final%20Report.pdf
SAS Season B page: https://statistics.gov.rw/statistical-publications/agriculture-environment/seasonal-agricultural-survey-season-b-2026
SAS Season B report: https://statistics.gov.rw/sites/default/files/documents/2026-09/SAS%202026B_Main%20report.pdf
SAS Season B summary: https://statistics.gov.rw/sites/default/files/documents/2026-09/SAS%202026B_Summary%20in%20English.pdf
SAS Season B workbook: https://statistics.gov.rw/sites/default/files/documents/2026-09/Tables_2026%20Season%20B.xlsx
SAS 2025 annual: https://statistics.gov.rw/sites/default/files/documents/2025-12/SAS%202025%20Final%20report.pdf
Microdata: https://microdata.statistics.gov.rw/index.php/catalog/124
Plot questionnaire: https://www.microdata.statistics.gov.rw/index.php/catalog/124/download/1114
PSTA 5: https://www.minagri.gov.rw/fileadmin/user_upload/Minagri/Publications/Policies_and_strategies/PSTA_5_Abridged_version_Final.pdf
NST2: https://www.minecofin.gov.rw/fileadmin/user_upload/Minecofin/Publications/STRATEGIES/NST_2/NST2_2024-2029_Abridged.pdf
Umushyikirano resolutions:
https://www.umushyikirano.gov.rw/fileadmin/user_upload/Umushyikirano/Imyanzuro/Resolutions_20th_UMUSHYIKIRANO_-_17_Feb_26.pdf
AHS 2024 report: https://statistics.gov.rw/sites/default/files/documents/2025-10/AHS%202024_Main%20report.pdf
AHS 2024 page: https://statistics.gov.rw/data-sources/surveys/Agricultural-Household-Survey/agricultural-household-survey-2024
CHIRPS: https://www.chc.ucsb.edu/data/chirps
RwaSIS: https://www.rab.gov.rw/1-1/news-details/rab-launches-rwanda-digital-soil-information-system-to-improve-agricultural-productivity
SAS Season A 2024, not the base file:
https://statistics.gov.rw/sites/default/files/documents/2024-10/SAS_2024%20Season%20A%20Final%20Report%20%282%29_compressed.pdf
GitLab: https://gitlab.com/mukunzijonathan/umusaruro
KT Press, 23 March 2026: https://www.ktpress.rw/2026/03/kagame-criticizes-slow-progress-in-agriculture-urges-shift-from-promises-to-action/
AllAfrica, 29 June 2026, news not Hansard: https://allafrica.com/stories/202606300076.html

7. Phases

Phase	Done when
1. Contract	Both people accept this document. No screen yet.
2. Data file	Second person checks every cell. Nationals reproduce. CHIRPS is computed.
3. Rules	Tests: Nyagatare, Kirehe, Kamonyi, Gakenke, Ngoma, one blank.
4. Django	Map, district page, method page. Source on every number.
5. Season B	Beside Season A. Cannot overwrite it. Table 29 footer resolved.
6. Submit	Public repo, live link, disclosure. 30 October 2026, 23:59.